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JOSÉ RIZAL UNIVERSITY

80 SHAW BOULEVARD
MANDALUYONG CITY

EXAMINATION BOOK

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DATE 10/06/2024

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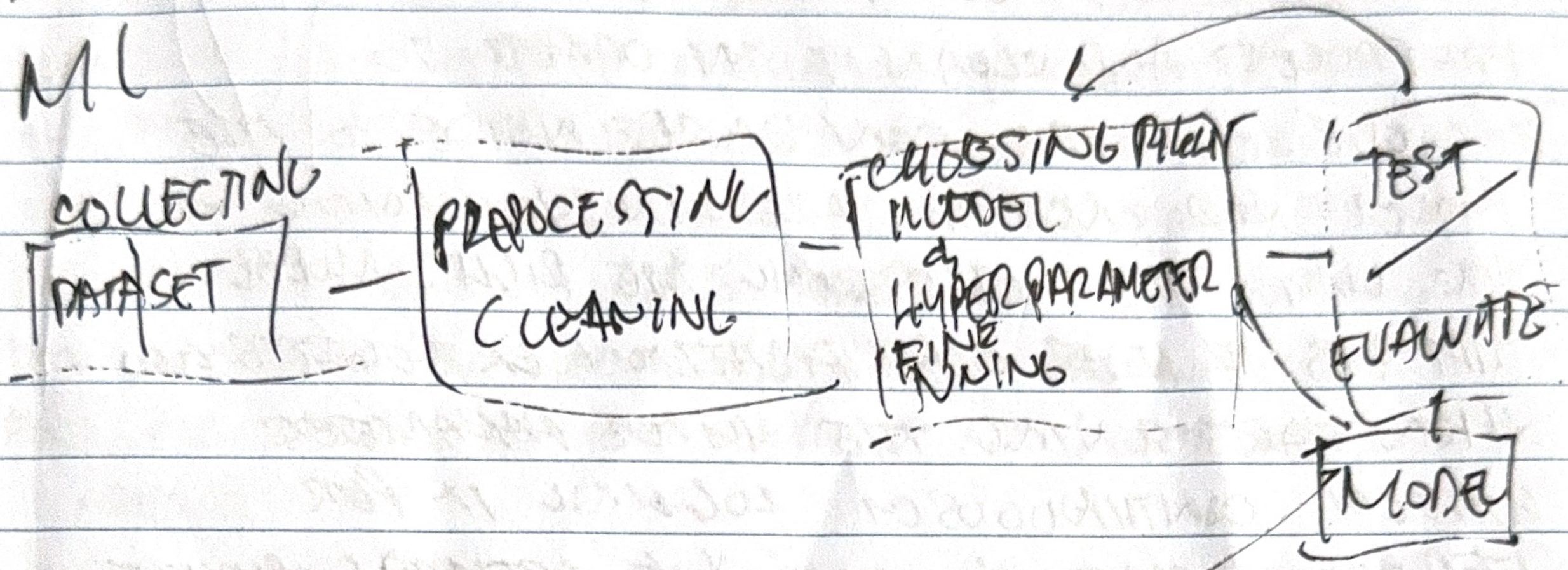
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EXAMINATION BOOK

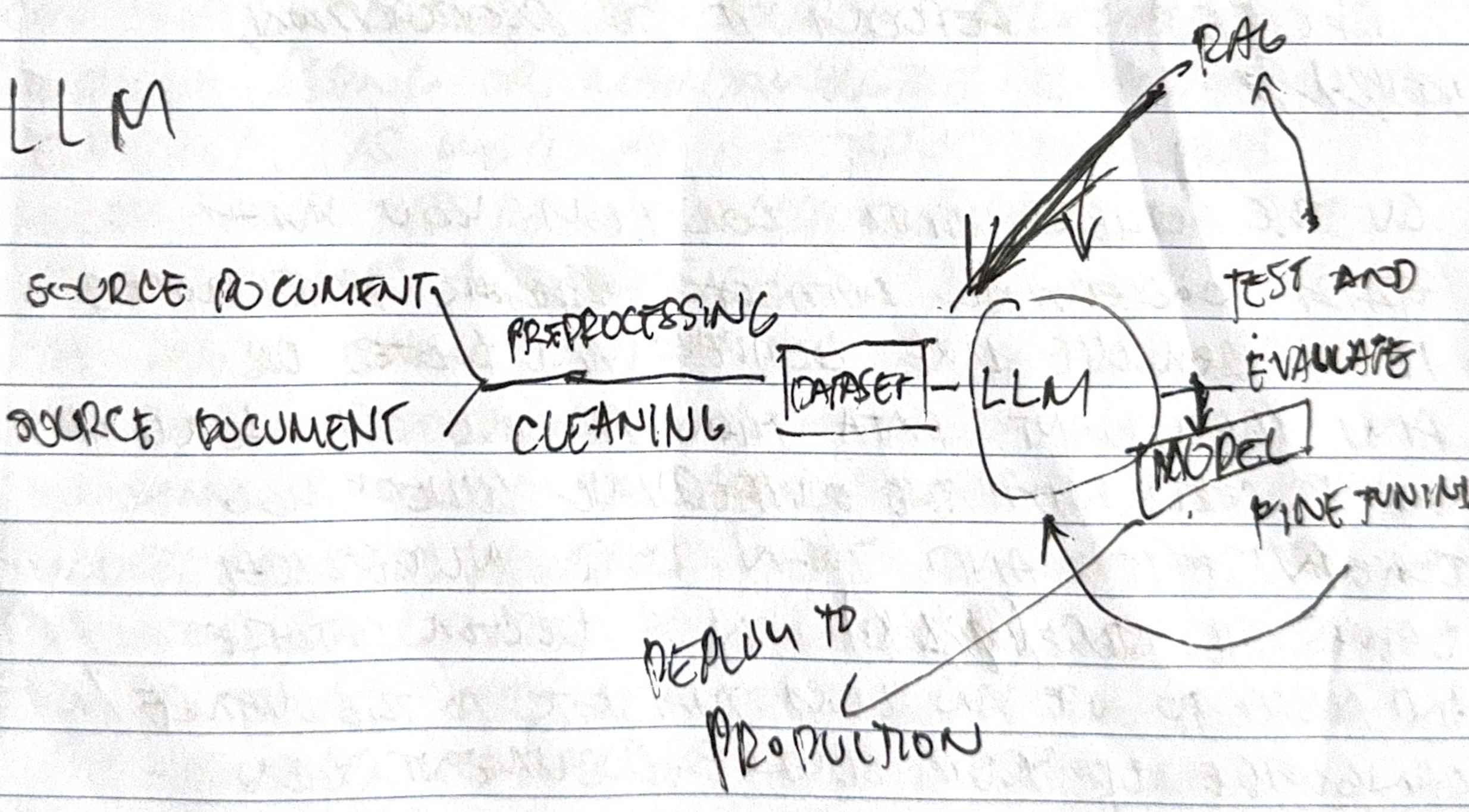
TEST I

1.

ML



LLM



FOR THE TRADITIONAL ML DEVELOPMENT YOU START WITH COLLECTING DATA AND PREPROCESS AND CLEAN IT IN ORDER TO REMOVE OUTLIERS CAN CAUSES NOISE TO THE MODEL AND ONCE YOU ARE FINISHED DOING SO, YOU CAN START CHOOSING THE RIGHT MODEL THAT FITS IF IT IS CLASSIFICATION OR REGRESSION THEN START TUNING THE HYPER PARAMETER WHICH CONTINUOUSLY LOOKING IN PER ITERATION TO FIND THE OPTIMAL VALUES AND EVALUATING THE RESULTS, IF THE MODEL IS ALREADY YIELDING A GOOD RESULTS TO THE TESTING YOU'VE DONE THATS HOW YOU CAN DECIDE TO DEPLOY IT TO PRODUCTION,

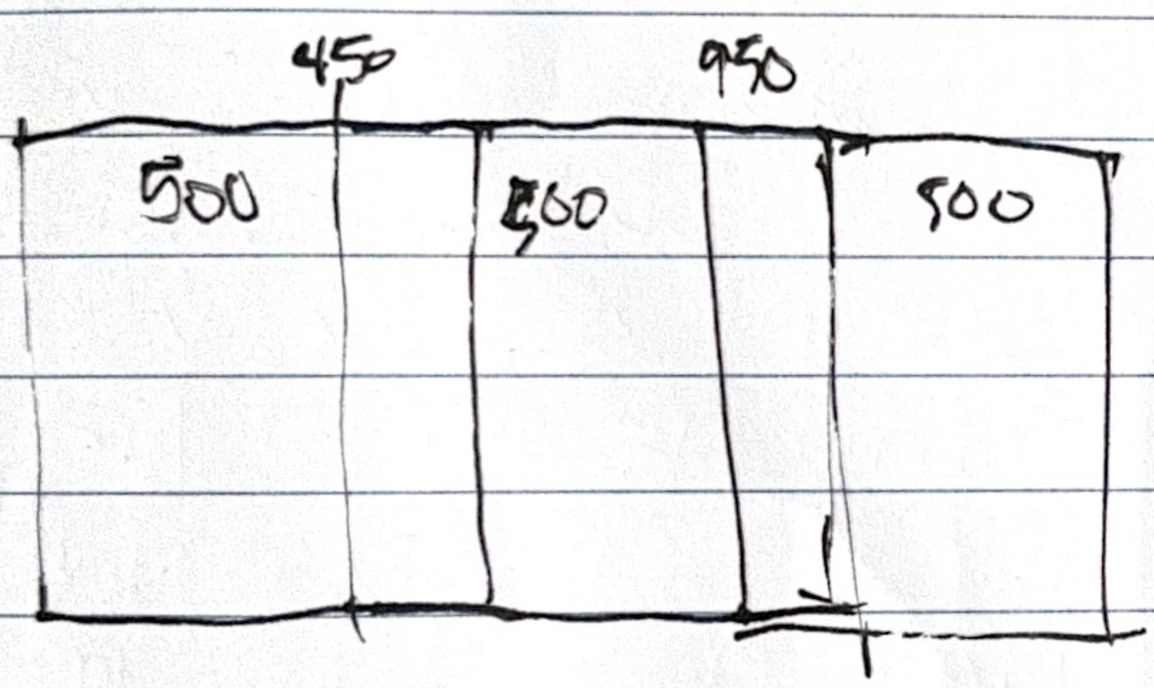
ON THE OTHER HAND, FOR LLM YOU MAY START COLLECTING DATASETS PREPROCESS IT TO REMOVE UKE LOGADERS AND ROOTED OR ANY IRRELEVANT DATA THEN PROCESS CHUNKING IT AND PROCEED WITH THE EMBEDDING WHERE YOU TOKENIZE IT AND TURN IT TO NUMERICAL VALUES THE STORE IN A VECTOR STORE AND CHOOSE TO USE AN LLM THAT FITS TO THE NATURE/ LANGUAGE OR YOUR SOURCE DOCUMENT THEN TEST AND EVALUATE BASED ON YOUR QUESTION THAT YOU SET, THEN CHECK IF IT REQUIRES FINE TUNING OR CHOOSING THE RAG ROUTE FLIP IT TO SATISFY THE REQUIREMENT BEFORE DEPLOYMENT.

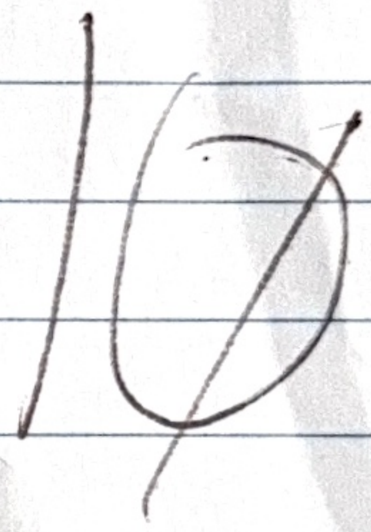
TEST II

1. THE MAIN OR ONE OF THE MAIN REASON WHY EFFECTIVE RAG DEVELOPMENT REQUIRES AN ITERATIVE PROCESS OF CHUNKING AND RETRIEVAL EVALUATION IS THERE IS NO ONE SIZE FITS ALL, AS YOU NEED TO FIND THE OPTIMAL VALUE FOR THE CHUNK SIZE AS WELL AS THE OVERLAP FOR THE CHUNKING PROCESS WHILE ADJUSTING THE TOP-K FOR THE RETRIEVAL MECHANISM, HENCE ENSURING TO FIND THE CHUNK SIZE AND OVERLAP PER ITERATION OPTIMAL

CAN ENSURE THAT IT CAN AID THE CHOSEN LLM ON A RAG-BASED MODEL / CHATBOT, AS THERE ARE LIMITATIONS ON HOW MUCH TOKEN IT CAN PROCESS & ENSURING AS WELL THAT THE CHUNKING HYPERPARAMETERS (SIZE & OVERLAP) CAN PROPERLY CHUNK THE SOURCE DOCUMENTS WITHOUT LOSING OR MAINTAINING THE QUALITY OF ITS CONTEXT. AS WELL AS THE TOP-K TO ENSURE THAT ONLY THE RELEVANT CHUNKS BASED ON THE USER QUERIES ARE RETRIEVED, AND THOSE ARE THE REASON WHY LEAVING IT UNCHANGED AFFECTS THE MODEL PERFORMANCE, AS IF YOU LEAVE IT UNCHANGED WHILE YOU ARE EVALUATING HOW THE MODEL RETRIEVES (ACCURACY) IT WILL NOT YIELD A GOOD RESULT OR A GROUNDED AND OR A GOOD WILL CAUSE THE LLM TO RETRIEVE ON ITS OWN PRE-TRAINED MEMORY OR VECTOR SPACE WHICH WILL CAUSE IT TO HALLUCINATE.

2. FIRST, YOU MUST CONSIDER IS TO TAKE NOTE OR ASSESS HOW LARGE IS THE SOURCE DOCUMENT. AND ON WHAT CATEGORY DOES IT FALL UNDER AND FROM THERE YOU CAN CONSIDER CHOOSING OR START WITH A BASELINE FOR THE CHUNK SIZE AND OVERLAP, ALSO OVERLAP IS NORMALLY 10% OF THE CHUNK SIZE SO THAT THE CONTEXT FOR EACH CHUNKS ARE PROPERLY PRESERVED WHILE YOU CONTINUOUSLY TEST AND EVALUATE THE RESULTS. AND LASTLY, THERE IS NO HARM IN CHANGING THE OVERLAP IF YOU NOTICE THAT THE CHUNK SIZE IS ALREADY ON ITS OPTIMAL VALUE THAT DOESN'T MEAN THAT EVEN IF IT'S 10% YOU CANNOT EXPERIMENT AND CHANGE IT TO 20% AND SO ON.





CONSIDER AS WELL THAT CHUNK SIZE IS SET USING CHARACTERS NOT TOKEN AND AS PER THIS VISUAL AID THIS IS HOW I VISUALIZE HOW CHUNKING AND OVERLAP WORKS.